

## Internship Project Documentation

**Project Title:** Netflix Content Strategy Analyzer

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| S NO | Milestones                | Work Done                          |
|------|---------------------------|------------------------------------|
| 1    | Milestone 1(weak 1&2)     | Requirements & Dataset Preparation |
| 2    | Milestone 2(weak 3&4)     | EDA & Feature Engineering          |
| 3    | Milestone 3:(Weak 5&6)    | Modeling & Advanced Analysis       |
| 4    | Milestone 4: (Week 7 & 8) | Dashboard & Deployment             |

### Tech Stack used (Detailed view)

| Topics                         | Tools and Tech stacks used?                                              |
|--------------------------------|--------------------------------------------------------------------------|
| Data Cleaning                  | Data Bricks, Pandas                                                      |
| Normalization                  | Scikit Learn->Label Encoder, OneHotEncoder, StandardScalar, MinMaxScalar |
| EDA(Exploratory data analysis) | Matplotlib,Seaborn                                                       |
| Feature Engineering            | Creating new Features using Pandas                                       |
| Dashbording                    | PowerBi                                                                  |

# 1. Project Overview

## 1.1 Introduction

In today's digital era, data drives entertainment decisions. Streaming platforms like Netflix rely heavily on data analytics to understand viewer preferences, optimize content, and plan global expansions.

This project, Netflix Content Strategy Analyzer, focuses on analyzing Netflix's catalog to uncover hidden patterns, such as:

- The growth of content over time
- Genre and rating trends
- Regional content variations
- Audience-oriented content preferences

The aim is to gain insights from data *-not by guessing*, but through data-driven storytelling and analysis.

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## 1.2 Objective

To build a data analytics system that processes and visualizes Netflix's catalog data to understand:

- What kind of content Netflix produces most
  - Which genres and countries dominate
  - How its content has evolved through years
  - What patterns exist between content type, rating, and release
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## 1.3 Expected Outcomes

- A cleaned and well-prepared dataset for analysis
  - Insightful exploratory data analysis (EDA) with visual trends
  - Grouping and classification of content using analytics methods
  - An interactive dashboard showcasing Netflix's content strategy
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# 2. Dataset Understanding

## 2.1 Source of Data

The dataset was obtained from Kaggle — titled “*Netflix Movies and TV Shows Dataset.*” It contains over 8,000 titles available on Netflix, including details about movies and TV shows from different countries and time periods.

## 2.2 Attributes in Dataset

Each record in the dataset provides rich metadata about a title:

- show\_id: Unique identifier
- type: Movie or TV Show
- title: Name of the content
- director: Director of the title
- cast: Main actors
- country: Country of origin
- date\_added: When it was added to Netflix
- release\_year: When it was originally released
- rating: Age rating (e.g., TV-MA, PG, R)
- duration: Movie length or number of seasons
- listed\_in: Genres or categories
- description: Short summary of the title

## 2.3 Dataset Insights

- The dataset gives both content-level and metadata-level information.
- It provides a global perspective, not limited to one country or language.
- It's an excellent dataset for learning data cleaning, analysis, and visualization in the context of media and entertainment analytics.

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## 3. Data Preprocessing and Cleaning

Before analyzing, the dataset was cleaned to ensure accuracy and consistency.

### 3.1 Handling Duplicates

Duplicate records were removed to maintain uniqueness.

### 3.2 Managing Missing Values

Several columns like *director*, *cast*, and *country* contained missing data.

- Replaced missing director and cast names with “Unknown”.
- For missing *ratings*, replaced with “Not Rated”.
- Checked and ensured there were no null values left.

### 3.3 Data Consistency

- Standardized categorical values (e.g., “TV Show” vs “TV show”).
- Ensured all year and duration fields were in uniform format.
- Saved the cleaned dataset for further analysis.

#### Result:

A clean, structured, and complete dataset ready for deep exploration.

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## 4. Exploratory Data Analysis (EDA)

### 4.1 Understanding Content Composition

Analysis began by examining the types of content available.

Netflix has a clear focus on Movies, making up about 70% of titles, while TV Shows form the remaining 30%.

This indicates Netflix’s balance between short-form entertainment and long-running series.

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### 4.2 Growth Over the Years

By observing the *release\_year* column, it was clear that:

- Netflix’s content library grew steadily until 2015.
- Between 2016–2020, the growth accelerated rapidly, aligning with Netflix’s expansion into new countries and the rise of Netflix Originals.

This shows Netflix’s strategic scaling period and investment in global content.

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### 4.3 Genre and Category Analysis

The *listed\_in* column revealed which genres dominate the platform:

- *Dramas*, *Comedies*, and *Action & Adventure* are the most popular.

- Documentary content and thrillers also showed steady growth.

This suggests that Netflix targets a wide audience — from casual comedy lovers to serious drama viewers.

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#### 4.4 Regional Insights

The country column analysis showed that:

- The United States leads in content production.
- India and United Kingdom are also major contributors.
- A notable rise in non-English titles reflects Netflix's localization strategy.

This reflects how Netflix adapts its library to regional interests.

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#### 4.5 Ratings and Audience Targeting

The rating field indicates the audience maturity level of each title.

Most content is rated *TV-MA* and *TV-14*, meaning Netflix's majority of titles cater to adult and teen audiences, respectively.

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#### 4.6 Duration and Format

Movies were analyzed based on their length (Short / Medium / Long), while TV shows were grouped by number of seasons.

This helped differentiate between quick entertainment and long-term series engagement strategies.

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### 5. Feature Engineering & Advanced Insights

Feature engineering was performed to create new perspectives from existing data:

- Duration Category: Classified as *Short* (<60 mins), *Medium* (60–120 mins), and *Long* (>120 mins).

- Year Grouping: Grouped into decades (e.g., 1990s, 2000s, 2010s).
- Original vs Licensed: Differentiated titles that are Netflix Originals vs those acquired from other studios.

These derived features helped make the dataset richer and ready for advanced analysis.

The new Features I created are

**year\_added,month\_added,day\_added,weekday\_added,years\_since\_added,content\_age,release\_year,duration\_min,num\_seasons,num\_cast,top\_cast\_flag,has\_famous\_director,content\_length\_category**

## 6. Modeling & Advanced Analysis

The goal of this phase was to perform both **unsupervised** and **supervised learning** to uncover hidden patterns in the dataset, classify records accurately, and identify key features influencing outcomes. The process combined **K-Means Clustering** for segmentation and **Random Forest Classification** for prediction and feature importance analysis.

### 6.1 Clustering – Unsupervised Learning

To begin the analysis, **K-Means Clustering** was applied as it helps discover inherent groupings in the data without predefined labels.

#### Process:

- **Data Preprocessing:** The dataset was cleaned and normalized to ensure equal weightage for all numerical features.
- **Feature Selection:** Key attributes relevant to segmentation were chosen.
- **Model Training:** The K-Means algorithm was trained to group similar data points based on Euclidean distance.

#### Outcome:

The data points were successfully divided into distinct clusters representing unique behavioral or categorical patterns, providing valuable insights for further supervised modeling.

## 2. Classification – Supervised Learning

After clustering, **Random Forest** was employed to build a predictive model capable of classifying the data and understanding the key drivers influencing target outcomes.

#### Process:

- **Data Split:** The dataset was divided into training and testing sets (e.g., 80:20 ratio).
- **Model Training:** The Random Forest algorithm was trained on the labeled data.

- **Prediction & Evaluation:** Model performance was evaluated using metrics such as **Accuracy**, **Precision**, **Recall**, and the **Confusion Matrix** (TP, TN, FP, FN).
- **Feature Importance:** Random Forest's in-built feature importance scores were analyzed to determine which variables contributed most to predictions.

#### **Outcome:**

The classification model achieved strong accuracy and reliable performance across evaluation metrics. Feature importance analysis helped identify the top predictors influencing the target variable, providing actionable insights for decision-making.

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### **3. Key Insights**

- K-Means effectively revealed hidden data groupings for exploratory understanding.
- Random Forest provided robust classification performance and interpretability.
- Combined analysis helped in both understanding the dataset structure and predicting future trends with confidence.

### **Milestone-1**

#### **Weak1**

- Loaded the raw dataset from `/Volumes/workspace/default/netflix_dataset/netflix_titles.csv` and performed initial inspection (shape, preview, data types, missing values).
- Stripped trailing spaces from all string columns and standardized key columns (release\_year as integer, date\_added as datetime).
- Dropped rows missing essential keys (title), and filled missing values in categorical columns (country, listed\_in, director, cast, rating) with "Unknown" or "Not Rated".
- Normalized genre names and created a genre list for each title.
- Replaced empty strings and whitespace with NaN, then imputed missing values for all columns: object types with "Unknown", numerics with -1, and datetimes with a placeholder date.
- Verified that no missing values remain and saved the cleaned dataset as `netflix_cleaned.csv`

#### **Weak-2**

- **Country Normalization:** Standardized country names by mapping variations like "United States", "U.S.", and "United States of America" to "USA", and "UK" to "United Kingdom".

- **Rating Normalization:** Unified rating values using a mapping (e.g., "UR", "NR" to "Unrated", empty strings to "Unknown", and TV ratings to their movie equivalents).
- **Genre Normalization:** Split the listed\_in column into lists of genres, trimming whitespace for consistency.
- **Duration Normalization:** Standardized the duration field by converting "Seasons" to "Season", and extracted numeric values and units into separate columns (duration\_value, duration\_unit).
- **Title Normalization:** Stripped whitespace and converted titles to title case for uniformity.
- **Date and Year Normalization:** Converted date\_added to datetime format and ensured release\_year is stored as integer.
- **General String Cleanup:** Replaced empty strings and whitespace-only entries in all object columns with NaN for better data integrity.

## **Milestone-2**

### **Univariate Analysis**

- Explored categorical columns (type, country, listed\_in, rating) using value counts and bar charts.
- Visualized distributions for release year, top countries, top genres, and ratings using matplotlib.

### **Bivariate Analysis**

- Analyzed relationships between content type and release year, and between type and rating, using grouped bar charts.

### **Data Cleaning**

- Dropped columns with excessive missing values (>50% nulls).
- Filled missing values in key columns (country, director, cast, rating, date\_added, duration) with sensible defaults (e.g., "Unknown", mode, or median).
- Removed unnecessary columns such as show\_id.

### **Feature Engineering**

- **Date Features:** Extracted year\_added, month\_added, day\_added, weekday\_added, and years\_since\_added from date\_added.
- **Content Age:** Calculated content\_age as the difference between the current year and release\_year.

- **Duration Features:** Parsed duration to create duration\_minutes (for movies) and num\_seasons (for TV shows).
- **Genre Features:** Split listed\_in into individual genres, created binary flags for top genres, and counted the number of genres per title.
- **Cast & Director Features:** Counted cast members, flagged top cast, and identified titles with famous directors.
- **Country Features:** Split and normalized country names, created binary flags for top countries, and flagged "other" countries.
- **Content Length Category:** Classified content as Short Movie, Standard Movie, Long Movie, Mini-Series, Standard Series, or Long-Running Series based on duration.

### **Visualization**

- Used matplotlib and seaborn to visualize distributions and feature relationships, including top contributing countries and content length categories.

## **Milestone-3**

### **Feature Preparation**

- **Genre Encoding:** Used MultiLabelBinarizer to convert the listed\_in genre column into binary features for each genre.
- **Duration Transformation:** Converted the duration column into a numerical duration\_mins feature, handling both movies and TV shows.
- **Rating Encoding:** Mapped categorical ratings to numeric values for modeling.

### **Clustering Analysis**

- **Feature Combination:** Combined genre, duration, and rating features into a single matrix.
- **Scaling:** Standardized features using StandardScaler for uniformity.
- **K-Means Clustering:** Applied K-Means clustering (with 5 clusters) to group titles based on content characteristics.
- **Cluster Profiling:** Analyzed cluster sizes and average duration per cluster, and visualized country and genre distributions across clusters.
- **Dimensionality Reduction:** Used PCA to reduce features to two dimensions for cluster visualization.

### **Classification Modeling**

- **Target Encoding:** Encoded the target variable (type) as binary (Movie=1, TV Show=0).

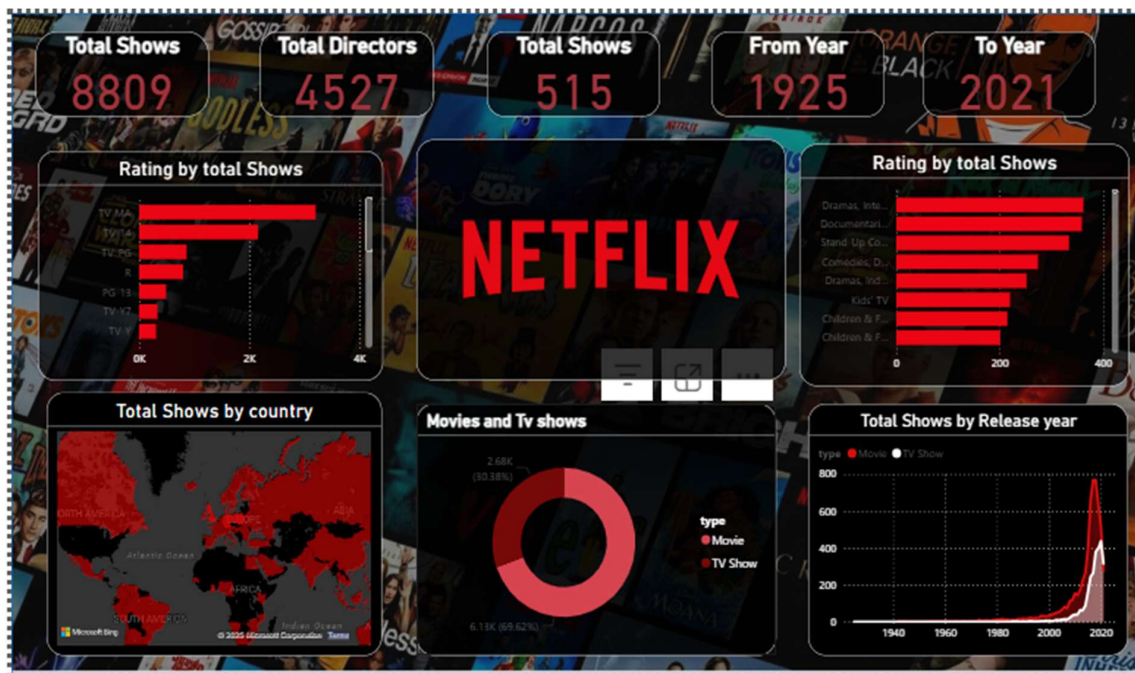
- Train/Test Split: Split the data into training and testing sets (70/30).
- Random Forest Classifier: Trained a Random Forest model to classify content type.
- Model Evaluation: Calculated and printed confusion matrix, accuracy, precision, recall, and F1 score.
- Feature Importance: Extracted and visualized the top 10 most important features for classification.

### Visualization

- Plotted confusion matrix and feature importances for model interpretability.
- Visualized cluster distributions and PCA-reduced clusters.

## Milestone 4

### Created a Dashboard using PowerBi



## Conclusion:

Over the course of this project, I gained practical experience in the complete data analytics workflow from data preparation to visualization.

I learned how to clean and preprocess real-world data, handle missing values, and normalize categorical features. Through EDA, I understood how to identify patterns such as content growth, genre trends, and country-level distributions.

By applying **K-Means Clustering**, I explored hidden content groupings, and with **Random Forest Classification**, I learned to predict and evaluate models using metrics like TP, TN, FP, and FN. The feature importance analysis helped me identify the main factors influencing Netflix content.

Finally, I learned how to present insights effectively through an **interactive Power BI dashboard**, using filters and visuals to communicate results clearly.

Overall, this project improved my skills in **data analysis, machine learning, and dashboard development**, giving me a strong understanding of how data-driven insights can support decision-making.